Lecture 25

Spinor tensor products and triality

Objectives. *Endomorphisms of spinors form a Clifford algebra, or its even part in odd dimension. As an orthogonal module this is a sum of exterior powers. In type D, chirality and duality decide which exterior powers occur.*

By the end of this lecture you will be able to:

- Identify $\mathrm{End}(S)$ equivariantly with Clifford elements and exterior powers.
- Prove $S \otimes S \cong \bigoplus_{k=0}^{n} \Lambda^k V$ for B_n.
- Prove the D_n rules for $S^{\pm} \otimes S^{\pm}$ and $S^{+} \otimes S^{-}$.
- Decompose the D_4 spinor products and explain triality.

25.1 Clifford algebras as endomorphism algebras

Let V be a complex vector space of dimension N with a nondegenerate symmetric bilinear form q, with the normalisations of Lecture 24. The Clifford algebra $\mathrm{Cl}(V)$ is generated by elements $c(v)$, linear in $v \in V$, with

$$c(u)c(v) + c(v)c(u) = 2q(u,v).$$

The skew maps $A_{uv} \colon V \to V$ are given by

$$A_{uv}(w) = q(v,w)u - q(u,w)v.$$

They span $\mathfrak{so}(V)$ and equal $\tfrac{1}{2}M_{u,v}$ from Lecture 24. The spin homomorphism $\sigma \colon \mathfrak{so}(V) \to \mathrm{Cl}(V)$ and its basic property, proved in Lecture 24, are

$$\sigma(A_{uv}) = \frac{1}{4}[c(u), c(v)], \qquad [\sigma(A), c(w)] = c(Aw). \tag{25.1}$$

The exterior model. Write $N = 2n$ or $N = 2n+1$. Choose $e_1, \dots, e_n, f_1, \dots, f_n \in V$ with

$$q(e_i, e_j) = q(f_i, f_j) = 0, \qquad q(e_i, f_j) = \frac{1}{2}\delta_{ij},$$

and, if N is odd, a vector u_0 orthogonal to them with $q(u_0, u_0) = 1$. Let $W = \mathrm{span}(e_1, \dots, e_n)$ and $S = \Lambda W$, so $\dim S = 2^n$. The algebra homomorphism $\rho \colon \mathrm{Cl}(V) \to \mathrm{End}(S)$ (called γ in Lecture 24) is

$$\begin{aligned} \rho(c(e_i)) &= e_i \wedge (\cdot), \\ \rho(c(f_i)) &= \text{signed contraction removing } e_i, \\ \rho(c(u_0)) &= (-1)^{\deg} \qquad (N \text{ odd}). \end{aligned}$$

The Cartan element $H_i = 2A_{e_i f_i}$ acts on V by

$$H_i e_j = \delta_{ij} e_j, \qquad H_i f_j = -\delta_{ij} f_j.$$

On S, Lecture 24 showed that $\rho(\sigma(H_i)) = N_i - \frac{1}{2}$, where N_i is 1 on wedges containing e_i and 0 otherwise. So the wedge e_T of the e_t with $t \in T$ has weight

$$\frac{1}{2}(s_1\varepsilon_1 + \cdots + s_n\varepsilon_n), \qquad s_i = +1 \ (i \in T), \quad s_i = -1 \ (i \notin T).$$

We use the positive roots $\varepsilon_i \pm \varepsilon_j$ $(i < j)$, with the ε_i added in type B. Lecture 24 proved:

- for B_n, $n \geq 2$, $S \cong L(\omega_n)$, where $\omega_n = \frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_n)$;
- for D_n, $n \geq 3$, $S = S^+ \oplus S^-$ with

$$S^+ = \bigoplus_{k \equiv n\,(2)} \Lambda^k W \cong L(\omega_n), \qquad S^- = \bigoplus_{k \equiv n-1\,(2)} \Lambda^k W \cong L(\omega_{n-1}),$$

  where $\omega_{n-1} = \frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_{n-1} - \varepsilon_n)$.

In the D_n convention, the top wedge $e_1 \wedge \cdots \wedge e_n$ has weight ω_n and degree n. Hence S^+ is the even-degree half only when n is even. Since s has $n - |T|$ minus signs,

$$\text{weights of } S^{\pm} = \Big\{\frac{1}{2}s : \ s_1 \cdots s_n = \pm 1\Big\}, \qquad \text{each with multiplicity one.}$$

For B_1, the spin operators satisfy

$$\rho\sigma(A_{e_1u_0})1 = \frac{1}{2}e_1, \qquad \rho\sigma(A_{f_1u_0})e_1 = -\frac{1}{2}.$$

Thus S is irreducible, with highest weight $\varepsilon_1/2 = \omega_1$.

An orthonormal basis. Put

$$u_{2j-1} = e_j + f_j, \qquad u_{2j} = i(e_j - f_j) \qquad (1 \leq j \leq n),$$

together with u_0 if N is odd. Then $q(u_a, u_b) = \delta_{ab}$. Let J be the index set, so $|J| = N$, and put $\gamma_a = c(u_a)$. Thus

$$\gamma_a\gamma_b + \gamma_b\gamma_a = 2\delta_{ab}, \qquad \sigma(A_{u_au_b}) = \frac{1}{2}\gamma_a\gamma_b \quad (a \neq b).$$

For $I = \{a_1 < \cdots < a_k\} \subseteq J$ put

$$\gamma_I = \gamma_{a_1} \cdots \gamma_{a_k}, \qquad \gamma_\varnothing = 1,$$

and define the spans

$$\mathrm{Cl}_k = \mathrm{span}\{\gamma_I : |I| = k\},$$
$$\mathrm{Cl}^0 = \mathrm{span}\{\gamma_I : |I| \text{ even}\},$$
$$\mathrm{Cl}^1 = \mathrm{span}\{\gamma_I : |I| \text{ odd}\}.$$

Lemma 25.1 (Reordering · *P*)**.** *Let $I, K \subseteq J$ and $a \in J$.*

(i) $\gamma_I\gamma_K = \pm\gamma_{I \triangle K}$*, where $\triangle$ is the symmetric difference.*
(ii) $\mathrm{Cl}(V)$ *is spanned by the γ_I; thus* $\mathrm{Cl}(V) = \mathrm{Cl}^0 + \mathrm{Cl}^1$.
(iii) $\gamma_a\gamma_I\gamma_a = (-1)^{|I|}\gamma_I$ *if $a \notin I$, and* $\gamma_a\gamma_I\gamma_a = (-1)^{|I|-1}\gamma_I$ *if $a \in I$.*

Proof. Distinct γ_a anticommute and $\gamma_a^2 = 1$. Sorting the factors of $\gamma_I\gamma_K$ costs a sign per transposition, and repeated indices cancel. This is (i). The γ_a generate $\mathrm{Cl}(V)$, since c is linear, so (i) gives (ii). For (iii), move the left γ_a through γ_I. It anticommutes with each factor γ_b, $b \neq a$, and commutes with γ_a. Finally, $\gamma_a^2 = 1$. ∎

Reordering makes Cl^0 a subalgebra.

Proposition 25.2 (Clifford algebra and $\mathrm{End}(S)\cdot P$). *(a) If $N = 2n$, then $\rho\colon \mathrm{Cl}(V) \to \mathrm{End}(S)$ is an isomorphism, and the γ_I, $I \subseteq J$, form a basis of $\mathrm{Cl}(V)$.*

(b) If $N = 2n+1$, then ρ restricts to an isomorphism $\mathrm{Cl}^0 \to \mathrm{End}(S)$, and the γ_I with $|I|$ even form a basis of Cl^0.

Proof. Let $\mathcal{I}$ be the set of all subsets of J in case (a), and of all even subsets in case (b). In both cases $|\mathcal{I}| = 2^{2n} = \dim \mathrm{End}(S)$, and $\mathcal{I}$ is closed under $\triangle$.

First, $\operatorname{tr}\rho(\gamma_I) = 0$ for nonempty $I \in \mathcal{I}$. If $|I|$ is even, choose $a \in I$. If $|I|$ is odd, we are in case (a). Since $|J| = 2n$ is even, $I \neq J$, and we choose $a \notin I$. In both cases Lemma 25.1(iii) gives $\gamma_a\gamma_I\gamma_a = -\gamma_I$. As $\gamma_a^{-1} = \gamma_a$,

$$\operatorname{tr}\rho(\gamma_I) = \operatorname{tr}\bigl(\rho(\gamma_a)\rho(\gamma_I)\rho(\gamma_a)^{-1}\bigr) = -\operatorname{tr}\rho(\gamma_I).$$

Next, Lemma 25.1(i) gives $\gamma_I\gamma_K = \pm\gamma_{I\triangle K}$ and $\gamma_I\gamma_I = \pm 1$. Hence

$$\operatorname{tr}\bigl(\rho(\gamma_I)\rho(\gamma_K)\bigr) = \begin{cases} \pm 2^n, & I = K, \\ 0, & I \neq K. \end{cases}$$

Suppose $\sum_{I\in\mathcal{I}} x_I\rho(\gamma_I) = 0$. Multiply by $\rho(\gamma_K)$ and take traces:

$$0 = \sum_{I\in\mathcal{I}} x_I \operatorname{tr}\bigl(\rho(\gamma_I)\rho(\gamma_K)\bigr) = \pm 2^n x_K.$$

So the 2^{2n} operators $\rho(\gamma_I)$ form a basis of $\mathrm{End}(S)$.

The γ_I, $I \in \mathcal{I}$, span $\mathrm{Cl}(V)$ in case (a) by Lemma 25.1(ii), and span Cl^0 in case (b) by definition. A spanning set with independent images is a basis, and ρ maps it bijectively onto a basis. ∎

25.2 Exterior powers and the odd-dimensional square

The algebra $\mathfrak{so}(V)$ acts on $\mathrm{Cl}(V)$ by $X \mapsto [\sigma(A), X]$. This commutator is a derivation of the associative product. By (25.1),

$$[\sigma(A), c(v_1)\cdots c(v_k)] = \sum_{i=1}^{k} c(v_1)\cdots c(Av_i)\cdots c(v_k). \tag{25.2}$$

By Lecture 14, $S \otimes S^* \cong \mathrm{End}(S)$, where A acts on $\mathrm{End}(S)$ by $T \mapsto [\rho(\sigma(A)), T]$. As ρ is an algebra homomorphism,

$$\rho\bigl([\sigma(A), X]\bigr) = [\rho(\sigma(A)), \rho(X)].$$

Each $\sigma(A)$ is a combination of the $\gamma_a\gamma_b$, so it lies in Cl^0 and its commutator preserves Cl^0. Hence Proposition 25.2 gives equivariant isomorphisms

$$S \otimes S^* \cong \mathrm{End}(S) \cong \begin{cases} \mathrm{Cl}(V), & N = 2n, \\ \mathrm{Cl}^0, & N = 2n+1. \end{cases} \tag{25.3}$$

Proposition 25.3 (Exterior powers in the Clifford algebra · *P*). *For $0 \le k \le N$ define*

$$\varphi_k(v_1 \wedge \cdots \wedge v_k) = \frac{1}{k!} \sum_{\pi \in S_k} \operatorname{sgn}(\pi)\, c(v_{\pi(1)}) \cdots c(v_{\pi(k)}),$$

and let $\Omega = \gamma_J$ be the product of all γ_a.

(a) φ_k is an $\mathfrak{so}(V)$-equivariant isomorphism $\Lambda^k V \to \mathrm{Cl}_k$.

(b) Left multiplication by Ω is an equivariant bijection $\mathrm{Cl}_k \to \mathrm{Cl}_{N-k}$. Hence $\Lambda^k V \cong \Lambda^{N-k} V$.

Proof. The sum is multilinear and alternating in $(v_1, \dots, v_k)$, so φ_k is defined on $\Lambda^k V$. Apply $[\sigma(A), \cdot\,]$ to each term and use (25.2). The result is φ_k of $\sum_i v_1 \wedge \cdots \wedge Av_i \wedge \cdots \wedge v_k$, so φ_k is equivariant. For distinct indices the γ's anticommute, so each reordered product equals $\operatorname{sgn}(\pi)\gamma_I$:

$$\varphi_k(u_{a_1} \wedge \cdots \wedge u_{a_k}) = \gamma_{a_1} \cdots \gamma_{a_k} = \gamma_I.$$

Thus φ_k maps a basis of $\Lambda^k V$ onto the spanning set of Cl_k.

For (b), Lemma 25.1 gives

$$\begin{aligned} \Omega\gamma_I &= \pm\gamma_{J\setminus I} && \text{by (i)},\\ \gamma_a \Omega \gamma_a &= (-1)^{N-1}\Omega && \text{by (iii)},\\ \Omega^2 &= \pm 1 && \text{by (i)}. \end{aligned}$$

The first line gives $\Omega\,\mathrm{Cl}_k \subseteq \mathrm{Cl}_{N-k}$. By the second, Ω commutes with each $\gamma_a\gamma_b$, hence with $\sigma(A)$, so left multiplication by Ω commutes with $[\sigma(A), \cdot\,]$. By the third, this map is bijective.

It remains to prove that the γ_I with $|I| = k$ are independent. For $N = 2n$ this is Proposition 25.2(a). For $N = 2n+1$ and k even it is Proposition 25.2(b). For $N = 2n+1$ and k odd, $\gamma_I = \pm\Omega\gamma_{J\setminus I}$ with $|J \setminus I|$ even, and multiplication by Ω is injective. Finally $\varphi_{N-k}^{-1} \circ \Omega \circ \varphi_k$ is an equivariant isomorphism $\Lambda^k V \to \Lambda^{N-k} V$. ∎

Proposition 25.4 (Duals of spin modules · *P*). *(a) For B_n, $S^* \cong S$ for every n.*

(b) For D_n with even $n \ge 4$, $(S^+)^* \cong S^+$ and $(S^-)^* \cong S^-$.

(c) For D_n with odd $n \ge 3$, $(S^+)^* \cong S^-$ and $(S^-)^* \cong S^+$.

Proof. The weights of a dual module are the negatives of the weights, with the same multiplicities. Negating $\frac{1}{2}s$ changes the sign product by

$$(-s_1) \cdots (-s_n) = (-1)^n\, s_1 \cdots s_n.$$

In type B every sign vector occurs once, so $\operatorname{ch} S^* = \operatorname{ch} S$. In type D, the sign product distinguishes S^+ from S^-. So $\operatorname{ch}(S^\pm)^*$ equals $\operatorname{ch} S^\pm$ for n even and $\operatorname{ch} S^\mp$ for n odd. Finite-dimensional modules with equal characters are isomorphic, by complete reducibility and highest-weight subtraction (Lecture 17). ∎

Theorem 25.5 (Spinor square for B_n · *P*). *Let $N = 2n+1$ with $n \ge 1$. Then*

$$S \otimes S \cong \operatorname{End}(S) \cong \bigoplus_{k=0}^{n} \Lambda^k V.$$

Proof. By Proposition 25.4(a), $S \otimes S \cong S \otimes S^*$. By (25.3) and Proposition 25.3(a),

$$S \otimes S^* \cong \mathrm{Cl}^0 = \bigoplus_{k \text{ even}} \mathrm{Cl}_k \cong \bigoplus_{\substack{0 \le k \le 2n+1 \\ k \text{ even}}} \Lambda^k V.$$

The middle sum is direct since the even γ_I form a basis. For even $k > n$, replace $\Lambda^k V$ by $\Lambda^{2n+1-k}V$, using Proposition 25.3(b). Each $j \in \{0, \dots, n\}$ now occurs once: as $k = j$ if j is even, and from $k = 2n + 1 - j$ if j is odd. ∎

25.3 Chirality and the D_n rules

From now on $N = 2n$ with $n \ge 1$. Define $S^\pm$ by the degree parities above, including $n = 1, 2$. The D_n highest-weight labels require $n \ge 3$. Let $\Gamma \in \mathrm{End}(S)$ act by $+1$ on S^+ and by -1 on S^-. By Proposition 25.2(a) we regard Γ as an element of $\mathrm{Cl}(V)$.

Lemma 25.6 (The chirality element · *P*). *We have $\Gamma^2 = 1$ and $\Gamma = \lambda\Omega$ for some $\lambda \in \mathbb{C}^\times$. The element Γ commutes with Cl^0 and anticommutes with Cl^1. Left multiplication by Γ is an equivariant bijection $\mathrm{Cl}_k \to \mathrm{Cl}_{2n-k}$.*

Proof. First, $\Gamma^2 = 1$, as Γ acts by ± 1 on $S^\pm$. Next, each $\rho(c(v))$ is a combination of the operators $e_i \wedge (\cdot)$ and of contractions. These change the degree by ± 1, so $\rho(\gamma_a)$ exchanges S^+ and S^-. Since $|J| = 2n$, Lemma 25.1(iii) gives $\gamma_a \Omega \gamma_a = -\Omega$. Hence

$$\Gamma\gamma_a = -\gamma_a\Gamma, \qquad \Omega\gamma_a = -\gamma_a\Omega.$$

Thus $\Gamma\Omega^{-1}$ commutes with every γ_a, hence with $\mathrm{Cl}(V) \cong \mathrm{End}(S)$, and is a nonzero scalar. Since Γ anticommutes with each γ_a, it commutes with even products γ_I and anticommutes with odd ones. The degree statement follows from Proposition 25.3(b), since $\Gamma = \lambda\Omega$. ∎

Left multiplication by Γ preserves Cl_n, and $\Gamma^2 = 1$. Put

$$\mathrm{Cl}_n^\pm = \{X \in \mathrm{Cl}_n : \Gamma X = \pm X\}, \qquad \Lambda_\pm^n V = \varphi_n^{-1}\big(\mathrm{Cl}_n^\pm\big).$$

These are submodules and $\Lambda^n V = \Lambda_+^n V \oplus \Lambda_-^n V$.

Theorem 25.7 (Chirality rules for D_n · *P*). *Let $N = 2n$ with $n \ge 3$. Then*

$$S^+ \otimes S^+ \cong L(2\omega_n) \oplus \bigoplus_{j \ge 1,\ n-2j \ge 0} \Lambda^{n-2j} V,$$

$$S^- \otimes S^- \cong L(2\omega_{n-1}) \oplus \bigoplus_{j \ge 1,\ n-2j \ge 0} \Lambda^{n-2j} V,$$

$$S^+ \otimes S^- \cong \bigoplus_{j \ge 0,\ n-1-2j \ge 0} \Lambda^{n-1-2j} V.$$

Moreover $\Lambda_+^n V \cong L(2\omega_n)$ and $\Lambda_-^n V \cong L(2\omega_{n-1})$.

Proof. Step 1: blocks. Put $P_\pm = \tfrac{1}{2}(1 \pm \Gamma)$, the projections onto $S^\pm$. For signs a, b, the space $\mathrm{Hom}(S^a, S^b)$ is $P_b \,\mathrm{End}(S) P_a$. It is a submodule isomorphic to $S^b \otimes (S^a)^*$, because $\sigma(A) \in \mathrm{Cl}^0$ commutes with Γ. By Lemma 25.6,

$$P_b X P_a = \begin{cases} \delta_{a,b}\, P_b X, & X \in \mathrm{Cl}^0, \\ \delta_{-a,b}\, P_b X, & X \in \mathrm{Cl}^1. \end{cases}$$

Hence $\operatorname{Hom}(S^a, S^b) = P_b \operatorname{Cl}^p$, where $p = 0$ if $a = b$ and $p = 1$ if $a \neq b$.

Step 2: degrees. Write Cl^p as the sum of the $\operatorname{Cl}_k \oplus \Gamma \operatorname{Cl}_k$ with $k < n$, $k \equiv p$, and of Cl_n if $n \equiv p$. Since $P_b \Gamma = b P_b$,

$$P_b\big(\operatorname{Cl}_k \oplus \Gamma \operatorname{Cl}_k\big) = P_b \operatorname{Cl}_k .$$

For $k < n$ consider

$$\operatorname{Cl}_k \longrightarrow P_b \operatorname{Cl}_k, \qquad x \longmapsto P_b x = \frac{1}{2} x + \frac{b}{2} \Gamma x.$$

It is injective, as $\Gamma x \in \operatorname{Cl}_{2n-k}$ and $2n - k \neq k$. It is equivariant, since Γ commutes with each $\sigma(A)$. Also $P_b \operatorname{Cl}_n = \operatorname{Cl}_n^b$. The pieces lie in different $\operatorname{Cl}_k \oplus \operatorname{Cl}_{2n-k}$, so

$$\operatorname{Hom}(S^a, S^b) \cong \bigoplus_{k<n,\ k\equiv p} \Lambda^k V \ \oplus \ \begin{cases} \operatorname{Cl}_n^b, & n \equiv p, \\ 0, & n \not\equiv p. \end{cases}$$

Step 3: duals. Put $\eta = (-1)^n$. By Proposition 25.4, $(S^c)^* \cong S^{\eta c}$ for each sign c. Hence

$$S^b \otimes S^b \cong S^b \otimes (S^{\eta b})^* = \operatorname{Hom}(S^{\eta b}, S^b),$$
$$S^+ \otimes S^- \cong S^+ \otimes (S^{-\eta})^* = \operatorname{Hom}(S^{-\eta}, S^+).$$

In the first line $p \equiv n$, since $\eta b = b$ exactly when n is even. In the second line $p \not\equiv n$. Step 2 gives the third formula with $k = n - 1 - 2j$, and

$$S^b \otimes S^b \cong \operatorname{Cl}_n^b \oplus \bigoplus_{k<n,\ k\equiv n} \Lambda^k V.$$

Step 4: the middle degree. By Proposition 25.3(a) and Lecture 23,

$$\operatorname{Cl}_n^+ \oplus \operatorname{Cl}_n^- = \operatorname{Cl}_n \cong \Lambda^n V \cong L(2\omega_{n-1}) \oplus L(2\omega_n).$$

The weights of $S^b \otimes S^b$ are $\frac{1}{2}(s + s')$ with $s_1 \cdots s_n = s'_1 \cdots s'_n = b$. In ε-coordinates the two extreme weights force

$$2\omega_n = (1, \dots, 1, 1) \implies s = s' = (1, \dots, 1, 1), \quad b = +1,$$
$$2\omega_{n-1} = (1, \dots, 1, -1) \implies s = s' = (1, \dots, 1, -1), \quad b = -1.$$

By Step 3, Cl_n^b is isomorphic to a summand of $S^b \otimes S^b$. Hence Cl_n^- contains no $L(2\omega_n)$, and Cl_n^+ contains no $L(2\omega_{n-1})$. Uniqueness of multiplicities gives $\operatorname{Cl}_n^+ \cong L(2\omega_n)$ and $\operatorname{Cl}_n^- \cong L(2\omega_{n-1})$. ∎

25.4 The D_4 spinor products

Take $n = 4$ and $V = \mathbb{C}^8$. Use the D_4 roots and fundamental weights of Lecture 23. Using Lecture 23 for the exterior powers, write

$$8_v = V = L(\omega_1), \qquad 8_s = S^+ = L(\omega_4), \qquad 8_c = S^- = L(\omega_3),$$
$$28 = \Lambda^2 V = L(\omega_2), \quad 56_v = \Lambda^3 V = L(\omega_3 + \omega_4), \quad 35_s = L(2\omega_4).$$

Theorem 25.7 with $n = 4$ gives

$$8_s \otimes 8_s \cong 35_s \oplus \Lambda^2 V \oplus \Lambda^0 V = 1 \oplus 28 \oplus 35_s,$$
$$8_s \otimes 8_c \cong \Lambda^3 V \oplus \Lambda^1 V = 8_v \oplus 56_v.$$

We confirm both by highest weights. Use the Weyl formula of Lecture 23, with Weyl vector $\rho_W = (3,2,1,0)$, $r = \rho_W$ and $a = \lambda + \rho_W$. The denominator is

$$\prod_{i<j}(r_i^2 - r_j^2) = (9-4)(9-1)(9-0)(4-1)(4-0)(1-0)$$
$$= 5 \cdot 8 \cdot 9 \cdot 3 \cdot 4 \cdot 1 = 4320.$$

Therefore

$$\dim L(2\omega_4) = \frac{7 \cdot 12 \cdot 15 \cdot 5 \cdot 8 \cdot 3}{4320} = \frac{151200}{4320} = 35, \quad a = (4,3,2,1),$$
$$\dim L(\omega_3+\omega_4) = \frac{7 \cdot 12 \cdot 16 \cdot 5 \cdot 9 \cdot 4}{4320} = \frac{241920}{4320} = 56, \quad a = (4,3,2,0).$$

The square. The weights of $8_s \otimes 8_s$ are $\frac{1}{2}(s+s')$ with $s_1 \cdots s_4 = s'_1 \cdots s'_4 = 1$. A coordinate is ± 1 where s and s' agree and 0 where they differ, and they differ in an even number of places. The dominant weights and multiplicities are

$$(1,1,1,1)\colon 1, \qquad (1,1,0,0)\colon 2, \qquad (0,0,0,0)\colon 8.$$

For $(1,1,0,0)$, $s = (1,1,x,y)$ and $s' = (1,1,-x,-y)$ with $xy = 1$.

The weight $(1,1,1,1) = 2\omega_4$ is maximal, so $L(2\omega_4)$ is a summand. Its complement Q satisfies

$$\dim Q = 64 - 35 = 29 > 8 = \dim(8_s \otimes 8_s)_0.$$

So some simple summand of Q has nonzero highest weight. Since $2\omega_4$ has multiplicity one, it is not a weight of Q, so this highest weight is ω_2. Hence $Q \supseteq L(\omega_2)$, of dimension 28. The remaining one-dimensional module is trivial, since $\mathfrak{g} = [\mathfrak{g},\mathfrak{g}]$.

The mixed product. For $8_s \otimes 8_c$, $s_1 \cdots s_4 = 1$ and $s'_1 \cdots s'_4 = -1$, so s and s' differ in an odd number of places. The dominant weights are

$$(1,1,1,0)\colon 1, \qquad (1,0,0,0)\colon 4.$$

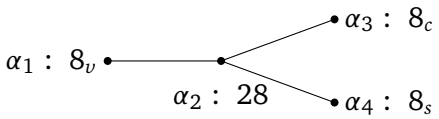
The maximal weight $\omega_3 + \omega_4$ gives a summand of dimension 56. The complement has dimension

$$64 - 56 = 8.$$

Its only possible dominant weight is ω_1, since 0 is not a weight and $\omega_3 + \omega_4$ has multiplicity one. So it is $L(\omega_1) = 8_v$.

25.5 Triality

The D_4 diagram displays the three outer nodes and their modules.



Every permutation of $\{1,3,4\}$ fixing 2 preserves the Cartan matrix.

Theorem 25.8 (Serre's theorem · *C*). *Let $\mathfrak{g}$ be complex semisimple with Cartan matrix $a_{ij} = \alpha_j(h_i)$ and generators $e_i \in \mathfrak{g}_{\alpha_i}$, $f_i \in \mathfrak{g}_{-\alpha_i}$, $h_i = [e_i, f_i] = \alpha_i^\vee$. Then $\mathfrak{g}$ is presented by these generators and the Serre relations stated in Lecture 10, whose coefficients are the entries a_{ij}. (Humphreys, Sections 18.3–18.4.)*

If a permutation π satisfies $a_{\pi(i)\pi(j)} = a_{ij}$, the elements $e_{\pi(i)}, f_{\pi(i)}, h_{\pi(i)}$ satisfy these relations. So there is a homomorphism $\tau_\pi\colon \mathfrak{g} \to \mathfrak{g}$ with $e_i \mapsto e_{\pi(i)}$, $f_i \mapsto f_{\pi(i)}$, $h_i \mapsto h_{\pi(i)}$. Since $\tau_{\pi^{-1}}$ inverts it on generators, τ_π is an automorphism.

Lemma 25.9 (Twisting · *P*). *For modules (M, ρ_M) and $(M', \rho_{M'})$, let M^π be M with x acting by $\rho_M(\tau_\pi^{-1}x)$. Then $(M \otimes M')^\pi = M^\pi \otimes (M')^\pi$, and*

$$L(\lambda)^\pi \cong L(\lambda^\pi), \qquad \lambda^\pi(h_i) = \lambda(h_{\pi^{-1}(i)}).$$

Proof. The tensor action $x \mapsto \rho_M(x) \otimes 1 + 1 \otimes \rho_{M'}(x)$ evaluated at $\tau_\pi^{-1}x$ is the tensor action of the twisted modules. The set of operators is unchanged, so $L(\lambda)^\pi$ is irreducible. Let v be a highest vector of $L(\lambda)$. In $L(\lambda)^\pi$, e_i acts by $e_{\pi^{-1}(i)}$, which sends v to 0. Also h_i acts on v by $\lambda(h_{\pi^{-1}(i)})$. The classification of Lecture 13 gives the isomorphism. ∎

For a fundamental weight the labels are

$$\omega_i^\pi(h_j) = \omega_i(h_{\pi^{-1}(j)}) = \delta_{i,\pi^{-1}(j)} = \delta_{\pi(i),j},$$

so $L(\omega_i)^\pi \cong L(\omega_{\pi(i)})$. Thus the permutations of $\{1,3,4\}$ permute the three eight-dimensional modules

$$8_v = L(\omega_1), \qquad 8_c = L(\omega_3), \qquad 8_s = L(\omega_4)$$

transitively. They fix ω_2, so 28 is fixed. This is *triality*.

Self-checks

(1) For D_5, which of $S^+ \otimes S^+$ and $S^+ \otimes S^-$ contains $\mathbb{C}$?

(2) List the summands and dimensions of $S \otimes S$ for B_3.

(3) In D_n, where does the copy of V occur, for n even and for n odd?

Answers. (1) $S^+ \otimes S^- \cong \Lambda^4 V \oplus \Lambda^2 V \oplus \mathbb{C}$, since $(S^+)^* \cong S^-$ for odd n. (2) $\mathbb{C} \oplus V \oplus \Lambda^2 V \oplus \Lambda^3 V$, with $1 + 7 + 21 + 35 = 64 = 8^2$. (3) In $S^+ \otimes S^-$ for n even, and in $S^\pm \otimes S^\pm$ for n odd.

Summary

The algebra $\mathrm{End}(S)$ is the Clifford algebra, or its even part in odd dimension. Hence $S \otimes S^*$ is a sum of exterior powers.

For B_n, S is self-dual and $S \otimes S \cong \bigoplus_{k=0}^{n} \Lambda^k V$. For D_n, the chirality element Γ sorts degrees by parity and splits the middle degree. Self-duality of $S^\pm$ depends on the parity of n. In D_4, triality permutes 8_v, 8_s and 8_c.

Reading

Fulton–Harris, §§20.1–20.3 (octonionic triality in §20.3). Goodman–Wallach, Chapter 6. Humphreys, §§18.3–18.4, for Serre's theorem.